Check for updates

LIMNOLOGY AND OCEANOGRAPHY
Letters

ASLO
Open Access

Limnology and Oceanography Letters 8, 2023, 229–246
© 2022 The Author. *Limnology and Oceanography Letters* published by Wiley Periodicals LLC on behalf of Association for the Sciences of Limnology and Oceanography.
doi: 10.1002/lol2.10294

CURRENT EVIDENCE

Understanding and predicting harmful algal blooms in a changing climate: A trait-based framework

Elena Litchman [1,2]*

[1]Kellogg Biological Station, Michigan State University, Hickory Corners, Michigan; [2]Department of Global Ecology, Carnegie Institution for Science, Stanford, California

Scientific Significance Statement

Harmful algal blooms (HABs) in both freshwater and marine systems present a significant problem for water quality and ecosystem services and may be increasing globally. Our ability to understand and predict HABs in diverse aquatic ecosystems is still limited. Trait-based approaches where the focus is on functional traits provide a mechanistic framework to better understand the environmental drivers of HABs. Characterizing trait differences between HAB-forming and other taxa, as well as across and within different HAB-forming species will help elucidate their ecological niches and adaptive potential under different conditions. Trait-based models should enable better HAB predictions at different temporal and spatial scales.

Abstract

The worldwide proliferation of harmful algal blooms (HABs) both in freshwater and marine ecosystems make understanding and predicting their occurrence urgent. Trait-based approaches, where the focus is on functional traits, have been successful in explaining community structure and dynamics in diverse ecosystems but have not been applied extensively to HABs. The existing trait compilations suggest that HAB taxa differ from non HAB taxa in key traits that determine their responses to major environmental drivers. Multi-trait comparisons between HAB-forming and other phytoplankton taxa, as well as within the HAB groups to characterize interspecific and intraspecific differences will help better define ecological niches of different HAB taxa, develop trait-based mechanistic models, and better identify environmental conditions that would likely lead to HABs. Building databases of HAB traits and using them in diverse statistical and mechanistic models will increase our ability to predict the HAB occurrence, composition, and severity under changing conditions, including the anthropogenic global change.

Harmful algal blooms (HABs) are a major problem in many freshwater and marine environments. The proliferation of toxic cyanobacteria, dinoflagellates, and other microalgae affects the structure and functioning of aquatic ecosystems worldwide (Heisler et al. 2008; Paerl and Huisman 2009). Excess biomass and subsequent oxygen depletion, toxins, lower nutritional quality for higher trophic levels, decrease in water quality, and other ecosystem alterations caused by HABs are increasing in many parts of the world. Increasing HABs are often attributed to global environmental change, especially

*Correspondence: elitchman@carnegiescience.edu

Associate editor: María J. González

Author Contribution Statement: EL conceived and wrote the paper.

Data availability statement: No new data have been used in this paper.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

warming, changes in land use and nutrient inputs (Huisman et al. 2018; Ho et al. 2019). Whether HABs in marine and freshwater ecosystems are increasing globally or only in some regions is debated (Hallegraeff et al. 2021; Wilkinson et al. 2022), but understanding the mechanisms producing HABs is critical. HAB-forming taxa belong to several taxonomic groups, and, according to the global IOC-UNESCO Taxonomic Reference List of Harmful Micro Algae (Moestrup et al. 2009), include at least 89 species of dinoflagellates, 35 cyanobacteria, 29 diatoms, 8 haptophytes, 6 raphidophytes, 3 dictyochophytes, and 2 pelagophytes. Cyanobacteria are the major bloom producers in freshwaters, and dinoflagellates dominate marine HABs, some diatoms and haptophytes also form blooms in coastal regions (Moestrup et al. 2009).

HABs negatively affect water quality and human well-being, with tremendous undesirable economic and health consequences (Anderson et al. 2000). Although we know a lot about the ecology of several major HAB taxa in marine and freshwater environments, our ability to predict blooms remains inadequate, especially on longer time scales or under rapidly changing conditions. Even relatively short-term, seasonal predictions of HABs using model ensembles may not work; for example, smaller blooms of shorter duration were predicted for Lake Erie for 2021 than were observed that year (Stumpf 2021). Similarly, correlational models of red tides do not perform well in highly dynamic coastal systems (McGowan et al. 2017). In general, statistical models of HABs mostly work on short-time scales, and process-based models are more suitable for longer-term predictions, especially under changing conditions (Ralston and Moore 2020).

Existing process-based models of HABs are often parameterized inadequately, with parameter values not based on the systematic compilations of many studies but on single studies of individual species (Liu et al. 2019), and thus may not predict the dynamics correctly, if different species or even different strains dominate. Moreover, it is hard to predict if the same toxic cyanobacteria would dominate in the future or they would be replaced by other taxa (Kokocinski et al. 2010; Thomas and Litchman 2016). The existing models focusing on the taxa that cause HABs at present may not be capable of forecasting the potential proliferation of different taxa in the future. There are models that predict blooms of the cyanobacterium *Microcystis* in Lake Erie (Rowe et al. 2016) but the blooms of other cyanobacteria cannot be predicted by those models, as they are species-focused. Whether the same or different HAB taxa would dominate in the changing climate is an important question because different species differ in their toxicity and the effects on food webs and biogeochemical cycling (Herrero and Flores 2008; Chorus and Welker 2021). Thus, trait differences across different HAB and non HAB taxa should be at the core of our efforts to understand and predict HABs at present and in the future.

We critically need novel frameworks that would increase our predictive capabilities for HABs (Wells et al. 2020). A trait-based perspective where the focus is on *functional traits* of species may become such a predictive mechanistic framework because it can identify and disentangle the myriad forces, such as abiotic drivers, trophic interactions, etc., structuring phytoplankton communities that include HAB taxa (Edwards et al. 2013*a*,*b*). By being species-agnostic, such a framework should be much more capable at predicting the dominance of novel taxa. Such a framework would help transform the ecology of HABs into a more predictive discipline, augmenting existing predictive tools, and should help address numerous fundamental and applied problems associated with HABs.

Surprisingly, despite the increased use of trait-based approaches to explain diverse patterns in ecological communities and species abundance, both in terrestrial and aquatic ecosystems (McGill et al. 2006; Litchman et al. 2007; Funk et al. 2017; Li et al. 2021), such approaches are only now starting to make inroads into the ecology of HABs (Kruk et al. 2017; Van de Waal et al. 2022). Here, I will introduce a trait-based framework and how it can be used to answer major questions about HABs and thus help understand and predict HAB dynamics. I will also discuss the challenges of using trait-based approaches. A trait-based perspective should improve our understanding of the mechanisms that drive HABs and thus benefit the science and management of aquatic ecosystems. The present work is not a comprehensive review of the trait information or the existing models of the HAB taxa, but rather a set of ideas for developing the mechanistic trait-based framework. These ideas should be applicable to HABs in both freshwater and marine environments, enabling cross-system comparisons that may enrich HAB ecology in general.

Overview of trait-based approaches

Using traits to explain community composition has become a mainstay of community ecology, especially for terrestrial plants (Wright et al. 2004; McGill et al. 2006; Lamanna et al. 2014). Major traits that explain variation in community composition have been identified in plants and other organisms (Díaz et al. 2016; Falster et al. 2017; Edwards and Steward 2018). The combinations of different traits and the trade-offs between them can define contrasting ecological strategies, such as fast growers vs good resource competitors, or the gleaner-opportunist continuum (Tilman 1990; Grover 1997).

Traits can be defined as "any morphological, physiological, or phenological heritable feature measurable at the level of the individual, from the cell to the whole organism, without reference to the environment or any other level of organization" (Garnier et al. 2016). Traits that affect fitness are termed *functional* traits (Violle et al. 2007). Traits can be morphological, physiological, life history, behavioral, as well as genomic and metabolic traits (Litchman and Klausmeier 2008; Litchman et al. 2021; Walker et al. 2022). In microbes, including phytoplankton, genomic and metabolic traits and the trade-offs may be important for inferring ecological strategies

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(Litchman et al. 2015*a*). Traits can be grouped according to different organismal functions, such as growth and reproduction, resource acquisition and avoidance of consumers and parasites/pathogens (Litchman and Klausmeier 2008) (Table 1). Traits can be classified as "response" and "effect" traits (Lavorel and Garnier 2002; Litchman et al. 2015*b*). *Response* traits characterize the responses of an organism to the environment and vice versa, *effect* traits describe the effects of an organism on the environment. A better characterization of response traits allows using them to understand and predict how organisms respond to changing environments. Some traits can be response and effect traits at the same time. For example, nitrogen (N) fixation is both a response trait (N-fixation increases under N-limitation) and an effect trait (N-fixation can increase N-availability in an ecosystem).

Relevant traits

The most important traits for predicting the dynamics of HAB taxa are the traits that characterize their responses to abiotic and biotic drivers and align with the major dimensions of their ecological niche. The main abiotic drivers for HAB taxa, as well as other phytoplankton, are nutrients (both macronutrients and micronutrients), light, temperature, and stability of the water column (Anderson et al. 1998; Litchman and Klausmeier 2008; Facey et al. 2019). pH may also be an important factor (Shapiro 1984; Raven et al. 2020). Biotic drivers include competition with other phytoplankton, zooplankton grazing, parasitism by viruses, fungi, and bacteria, and mutualistic interactions with bacteria and other phytoplankton (Work and Havens 2003; Hoke et al. 2021; Papoulis et al. 2021).

Consequently, key traits are those that describe how abiotic and biotic factors affect fitness (Table 1). They include cell size, nutrient- and light-utilization traits, temperature-response traits (Fig. 1), susceptibility to grazing, with cell/colony size often being the proxy for that, buoyancy/motility, the ability to fix atmospheric nitrogen, toxicity, and others (Table 1) (Litchman and Klausmeier 2008; Litchman et al. 2021).

The focus on trait differences among phytoplankton has a rich history. The seminal work of Colin Reynolds in freshwaters (Reynolds 1984, 1997; Reynolds et al. 2002), Ramon Margalef in marine environments (Margalef 1978), and others (Smayda 1997; Smayda and Reynolds 2001) described many morphological, physiological and life history traits and their associations in major taxonomic groups of phytoplankton. Although the modern trait-based framework continues this tradition, it became much more quantitative, comparing multiple quantitative traits and their distributions along environmental gradients (Edwards et al. 2012; Thomas et al. 2012; Mousing et al. 2018), similar to such approaches in terrestrial plants (Wright et al. 2004; Butler et al. 2017). Moreover, there is more focus on the eco-physiological traits directly connected to fitness (functional traits) that can be used to parameterize mechanistic models. Morphological traits, which are easier to characterize can sometimes be used to infer the eco-physiological traits (Kruk et al. 2010).

Recently, significant progress has been made in developing trait-based ecology of both marine and freshwater phytoplankton (Litchman and Klausmeier 2008; Schwaderer et al. 2011; Edwards et al. 2013*a*, 2016, Weithoff and Beisner 2019; Kruk et al. 2021). One of the major advances is the demonstration that traits can explain diverse occurrence patterns of different phytoplankton. For example, several eco-physiological traits measured in the lab, such as nutrient affinity, light utilization traits, and maximum growth rates could explain more than 80% of temporal variation in abundances of phytoplankton species in a marine ecosystem (Edwards et al. 2013*a*). Similarly, for freshwater phytoplankton we found that light utilization traits and maximum growth rates predicted how species abundances vary along light and nutrient gradients on a continental scale (Edwards et al. 2013*b*). The maximum growth rates and nutrient competitive abilities inferred from morphological traits explained seasonal succession in a shallow hypertrophic lake (Segura et al. 2013).

Despite these advances, HAB research rarely applies trait-based approaches, where the information on many traits and their variation across taxa and along environmental gradients is synthesized and used to explain species distributions and community structure. Moreover, trait-based models that ask what trait combinations would be selected for in given environments as providing the highest fitness (Follows et al. 2007; Falster et al. 2017; Klausmeier et al. 2020*a*) have not been applied to HABs. I propose that such empirical and modeling approaches can be used systematically to better understand the mechanisms that lead to HABs in different environments. I formulate several major questions and discuss how they can be answered using traits. Finally, I propose that a trait-based framework can be used to predict HABs in the future, including under rapidly changing conditions and on longer time scales.

Comparing traits between HAB and non-HAB taxa

One of the major questions is whether and how HAB taxa differ in their eco-physiology from other phytoplankton. Answering this question will help determine the niche differences between HAB and non-HAB taxa and, consequently, the conditions leading to HABs, and requires a quantitative comparison of the relevant eco-physiological traits, with a comprehensive coverage of those groups (Fig. 3). Existing trait compilations allow us to make some inferences, but as far as we know, no extensive comparison that focuses specifically on HAB taxa vs. other groups exists. Here I provide examples of relevant trait comparisons and highlight the gaps.

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Major types of functions, the associated traits and trait examples important for HAB taxa and other phytoplankton (not an exhaustive list). Trait classification modified from Litchman and Klausmeier (2008) and Litchman et al. (2021). Some representative references for select traits are also provided. Functions/traits in bold are often included in models of HABs.

Trait type	Function/trait(s)	Example traits
Genomic	Genome size and structure	Large (relative or absolute) genomes in some HAB taxa (Kudela et al. 2010; Larsson et al. 2011; Hong et al. 2016), smaller genomes, fewer transposable elements in toxic *Microcoleus* strains (Tee et al. 2021)
	GC content	Lower GC content in toxic *Microcoleus* strains (Tee et al. 2021)
	Presence of certain genes, metabolic pathways (e.g., toxin synthesis, defense genes)	Microcystin synthetase gene *mcy* (copy number, expression/transcript level) (Hellweger et al. 2022), viral defense genes (Morimoto et al. 2019), domoic acid synthesis genes *dab* (Brunson et al. 2018)
Morphological	Cell size, shape	Diameter (spherical cells), length, width, cell volume, aspect ratio (Ryabov et al. 2021)
	Coloniality	Colony size, filament length
Metabolic/physiological	**Nutrient (N, P, Fe, etc.)-dependent growth**	Maximum growth rate μ_{max}, half-saturation constant for growth k_s, nutrient affinity for growth
	Nutrient uptake and storage	Max resource acquisition rate, V_{max}, uptake affinity, half-saturation constant, k_s, storage capacity Q_{max}
	Light-dependent growth	Maximum light-dependent growth, light affinity α (initial slope), optimum irradiance I_{opt} (susceptibility to photoinhibition)
	Photosynthesis	Maximum light-dependent photosynthesis, light affinity α (initial slope)
	Respiration	Max respiration, environmental controls
	Environmental factor tolerance (**temperature**, pH, salinity, turbulence, toxins, etc.)	Optimum temperature for growth (Thomas and Litchman 2016) T_{opt}, thermal niche width, min and max temperatures for growth T_{min} and T_{max}, photosynthesis, respiration, and so on
	Nitrogen fixation (in cyanobacteria)	Presence/absence, maximum rate, density of heterocytes (as proxy of N fixation (De Tezanos Pinto and Litchman 2010), dependence on environmental factors
	Mixotrophy (e.g., in dinoflagellates)	Presence/absence, max ingestion rate, prey range, and so on
	Toxicity	Presence/absence, dependence on environmental factors (nutrients, light, etc.)
Life history/behavioral	Dormancy/resting stages	Akinete presence/number, environmental controls of dormancy (Wyman and Fay 1986)
	Motility/buoyancy	Presence/absence of flagella, gas vesicles (Fogg and Walsby 1971), swimming speed, environmental controls (Lewis et al. 2006)
	Sexual/asexual reproduction	Presence/absence, dependence on environmental factors (in dinoflagellates, Figueroa et al. 2015; diatoms, Annunziata et al. 2022)
	Defense strategy	Colony formation (Yang and Kong 2012, Zhang et al. 2021), cell shape change

Freshwater comparisons

Among the freshwater phytoplankton, cyanobacteria, major HAB-producing group, appear to have intermediate growth rates, high phosphorus uptake affinity but low nitrogen uptake affinity, likely influenced by the nitrogen fixers among the cyanobacterial species (Edwards et al. 2012). Cyanobacteria also have high initial slopes of the growth-irradiance curves α and low optimum irradiances for growth I_{opt} compared to other freshwater taxa, making them adapted to low light environments (Schwaderer et al. 2011). Interestingly, diatoms also tend to have high α, but also high I_{opt}, which makes them adapted to fluctuating light (Schwaderer et al. 2011).

Freshwater cyanobacteria can regulate their position in the water column through collapsible gas vesicles (Klemer 1983). This provides an advantage in accessing the optimal light and

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

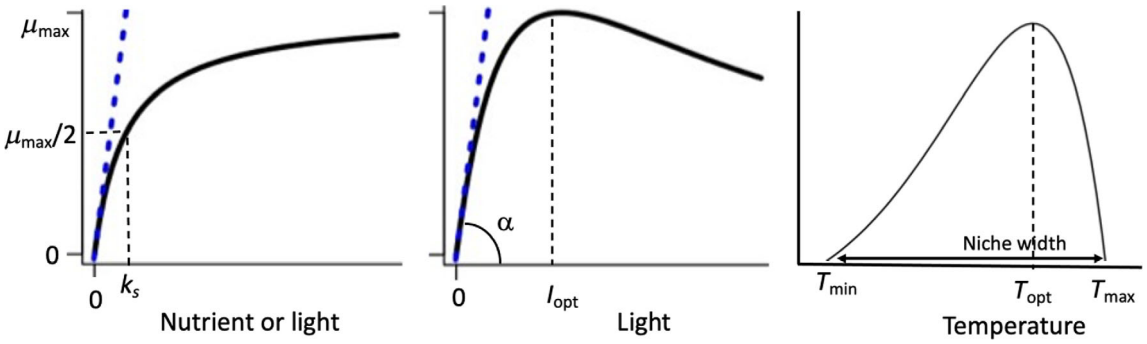


Fig. 1. Examples of key functional traits of HAB taxa and other phytoplankton. (**a**) Typical relationship between growth rate and resource (nutrient or light) level and the associated traits, the maximum growth rate μ_{max}, the growth affinity (initial slope of the growth-resource curve), and the half-saturation constant for growth k_s. (**b**) Light-dependent growth curve with photoinhibition, the traits are the maximum growth rate, the light affinity α (slope), the optimum irradiance I_{opt}, above which photoinhibition occurs. (**c**) Typical thermal performance curve (TPC) (growth dependence on temperature) and the associated traits, the optimum temperature for growth T_{opt}, the minimum and maximum temperatures where growth is non negative, T_{min} and T_{max}, respectively, and the thermal niche width.

nutrient conditions (Reynolds 2006). Interestingly, in marine environments many HAB-forming taxa can also adjust their position in the water, but mostly through active swimming using flagella. However, there are many taxa that possess the ability to regulate their position in the water column, either through regulating buoyancy or active swimming, that do not form HABs.

Freshwater cyanobacteria appear to have high optimum temperatures for growth compared to other groups but this difference may only be significant for temperate strains (Thomas et al. 2016). In the tropics, the temperature-related traits are similar across major groups (e.g., cyanobacteria, diatoms and green algae in freshwater have similar T_{opt}), possibly because of the fewer distinct thermal niches due to small temperature variation (Thomas et al. 2016). This dependence of trait differences on the strain isolation latitude may lead to different predictions of how warming would influence freshwater cyanobacterial blooms in temperate vs. tropical environments: rising temperatures may stimulate cyanobacterial blooms in temperate latitudes but not in the tropics, because of the similar temperature traits of different phytoplankton groups in low latitudes. This hypothesis has not been investigated but may be important for assessing the role of climate change in HAB proliferation in different geographic regions. A recent metaanalysis showed that the population dynamics of the bloom-forming cyanobacterium *Raphidiopsis* (*Cylindrospermopsis*) *raciborskii* strongly depended on temperature in temperate regions but only weakly in a tropical lake (Recknagel et al. 2019), thus supporting our trait-based hypothesis. In the tropics, cyanobacterial HABs can occur for prolonged periods of time, as they are less constrained by temperature changes (Zohary and Robarts 1990; Roegner et al. 2020). Tropical strains from major groups can still differ in other traits, hence, changes in other environmental factors may alter bloom severity and duration in the future.

Marine comparisons

In marine environments, dinoflagellates are one of the main taxonomic groups that produce HABs, known as "red tides." Their key functional traits differ significantly from other marine groups. The maximum growth rates of dinoflagellates are the lowest of most other marine phytoplankton groups (Edwards et al. 2012). Their scaled nitrogen and phosphorus uptake affinities are also lower than in other groups, making them inefficient nutrient competitors (Edwards et al. 2012) (Fig. 4C). It is likely that mixotrophy and motility help offset this inefficiency through enabling improved access to nutrients. Dinoflagellates also tend to have low initial slopes α of the growth-irradiance curves, which makes them poor light competitors (Edwards et al. 2015) (Fig. 4C). Interestingly, this is different from the major HAB-forming group in freshwaters, the freshwater cyanobacteria, which are good light competitors. The ability of dinoflagellates to migrate vertically to gain access to optimal light conditions may similarly offset their poor light-competitive abilities.

Similar to the freshwater groups, the differences in temperature optima for growth across major marine groups appear greater in higher latitudes, compared to the tropics (Thomas et al. 2016). This may mean that rising temperatures are more likely to preferentially stimulate dinoflagellate blooms in temperate climate, and less so in the tropics.

Multitrait comparisons

Considering the ranges of the observed values for multiple traits simultaneously shows remarkable differences across major taxa in a multitrait space for both marine and freshwater groups (Fig. 4). Note that the cuboids in Fig. 4 result from combining the ranges of individual traits for each taxonomic group and, therefore, illustrate the possible regions of trait space, which may not all be filled by existing species. Due to the trade-offs among traits, some trait combinations may not

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

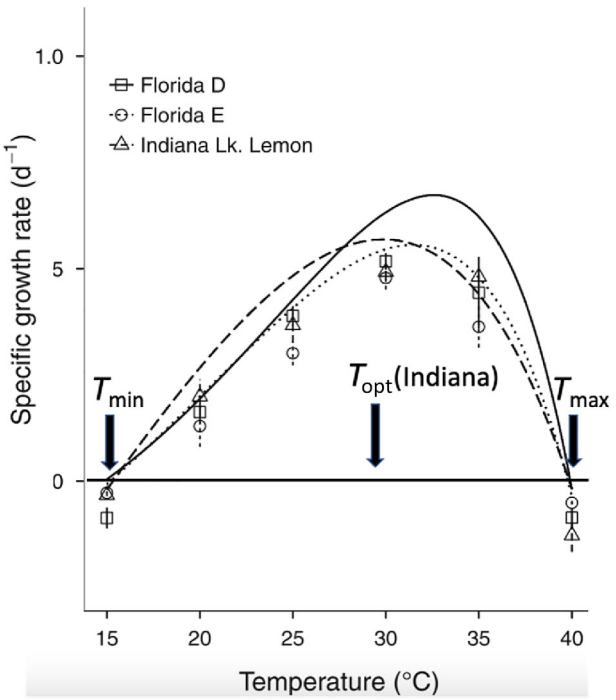


Fig. 2. Thermal performance curves of three strains of the HAB-producing cyanobacterium *Raphidiopsis* (*Cylindrospermopsis*) *raciborskii* and the associated traits, T_{min} (°C, lower thermal bound of positive growth rate), T_{opt} (°C, temperature at which growth rate is the highest), and T_{max} (°C, upper thermal bound of positive growth rate). Note the lower T_{opt} for the Indiana strain (T_{opt} [Indiana]) compared to the Florida strains (from Thomas and Litchman 2016).

be possible, resulting in some parts of the cuboids being empty (Litchman et al. 2007; Edwards et al. 2011). Individual species align along the trade-off surfaces within this volume (Edwards et al. 2011). We found that a three-way trade-off leads to a plane that includes both the marine and freshwater species' traits, so one can envision points representing trait combinations of individual species or strains aligning on such a plane (Edwards et al. 2011). With more data, individual species or strains can be represented in a multitrait space, instead of the individual trait ranges delineated by the cuboids.

Trade-offs and trait inference

Trade-offs among traits (pairwise and higher-dimension) are important in defining diverse ecological strategies. For example, in the compilation of light traits, we can identify taxa that are good light competitors but sensitive to photoinhibition, with high alpha and low I_{opt} = low light-adapted strategy vs. the high light-adapted strategy (Schwaderer et al. 2011). Toxicity also trades off with other traits, such as light competitive abilities (Kardinaal et al. 2007) or the ability to neutralize hydrogen peroxide (H_2O_2), one of the reactive oxygen species (ROS) produced by phytoplankton (Diaz and Plummer 2018; Hellweger et al. 2022). Characterizing trade-offs is also useful for inferring the missing traits, where the relationships between traits (often representing trade-offs) can be used to derive the unknown trait values when other traits are known.

Currently, many ecologically important traits, such as the resource-dependent or temperature-related traits, are not known even for key HAB species, so some traits need to be inferred. There are several methods to infer missing traits but all of them have limitations (Bruggeman 2011; Edwards et al. 2011; Penone et al. 2014). A better approach is to determine traits directly, through measurements (high amount of

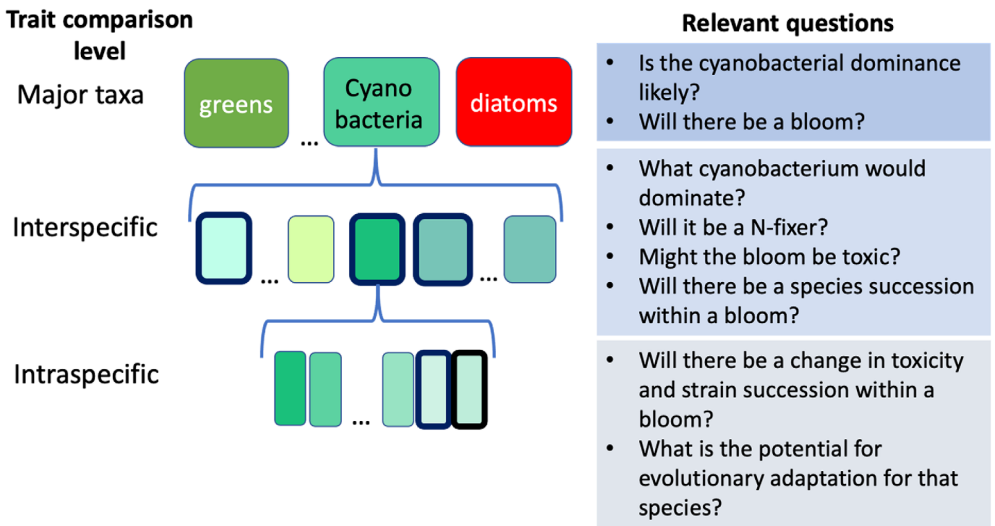


Fig. 3. Different levels of trait comparisons (across different taxonomic groups, species, and intraspecific) and examples of possible questions that can be addressed through trait comparisons at each level. At the interspecific and intraspecific levels, individual boxes represent different species and genotypes, respectively. The thick borders of some boxes symbolize toxin-producing species or strains.

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

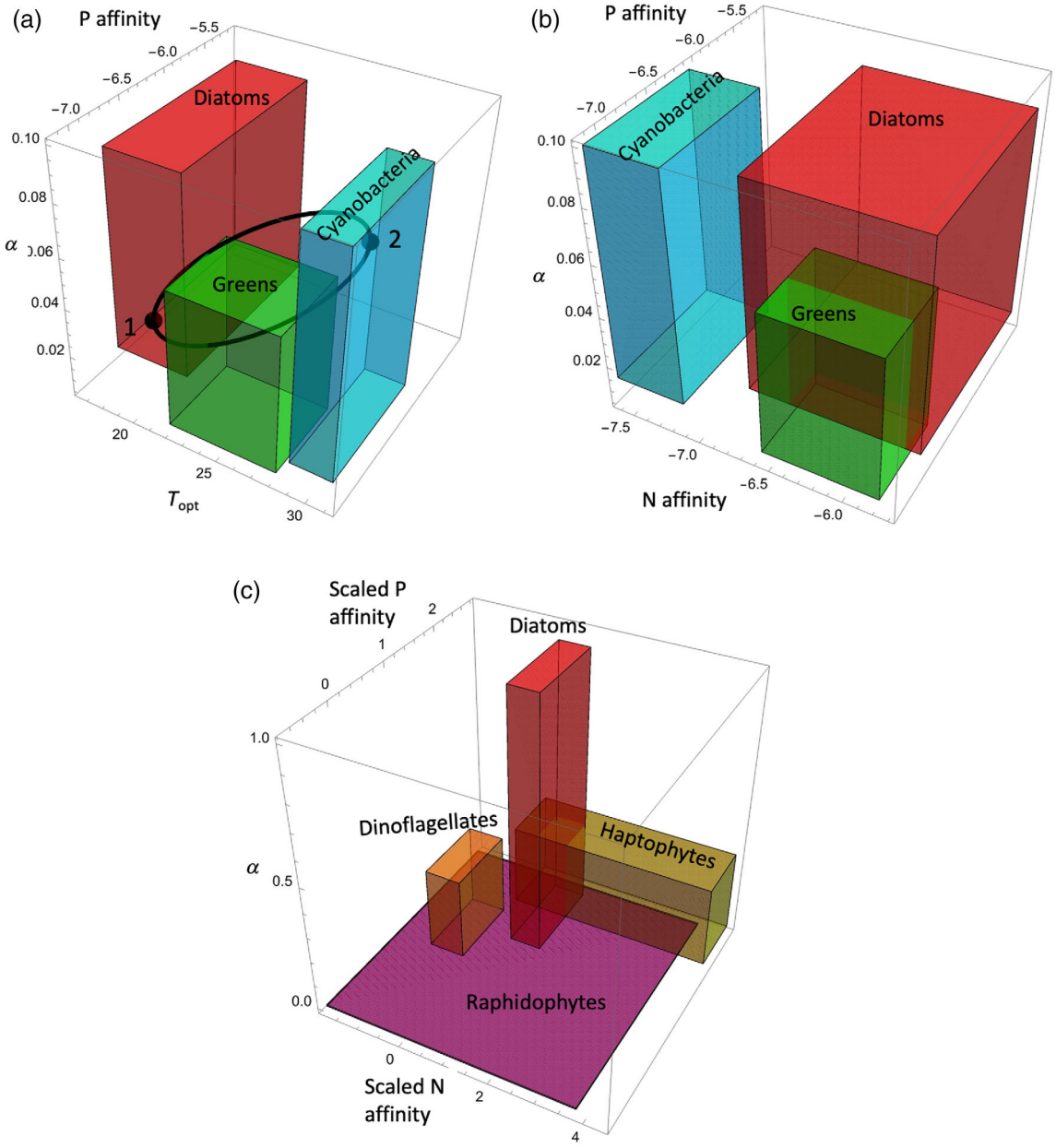


Fig. 4. Trait ranges for different taxonomic groups taken from published trait compilations. (**a**,**b**) Freshwater groups. The traits are N and P affinity ($\log_{10}$ L cell^{-1} d^{-1}), the affinity to use nitrogen and phosphorus, respectively, ranges (trait mean ± 95% CI) taken from Edwards et al. (2012), α is the initial slope of the growth-irradiance curve (μmol quanta^{-1} m^2 s d^{-1}), values show ranges taken from Schwaderer et al. (2011) and the optimum temperature for growth, T_{opt} (°C) (mean ± 95% CI) (from Thomas et al. 2016). (**c**) Marine groups. The traits are *scaled* N and P affinity ($\log_{10}$ L μmol^{-1} d^{-1}), the affinity to use nitrogen and phosphorus, respectively, normalized by cell volume (trait mean ± 95% CI) taken from Edwards et al. (2012) and α (μmol quanta^{-1} m^2 s d^{-1}) is the initial slope of the growth-irradiance curve normalized by temperature and daylength, values (ranges) taken from Edwards et al. (2015). Note the slightly different units for marine and freshwater groups. The circle and points in (**a**) represent the possible trajectories of trait combinations and specific combinations that may be selected for in different lakes or different seasons. Point 1: Low T_{opt}, α, and P_{aff} may be selected for in the beginning of the growth season, when water temperatures are low, and light and nutrients are high; over the course of the season, the optimal trait combinations change (shown as dark trajectories) and can reach point 2: this combination of traits may be advantageous later in the season, when temperatures are higher, but both P and light availability decline, favoring groups, species, or strains with high T_{opt}, α, and P_{aff}.

effort), and possibly, infer traits from the environmental data using the artificial intelligence approaches (*see* "Future directions" section). Different taxa can then be compared using multivariate statistical techniques, and their trait differences visualized in the multidimensional trait space (e.g., nonmetric multidimensional scaling, Chuckran et al. 2021).

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

This very broad comparison of traits of major taxonomic groups that include HAB taxa in freshwater and marine environments suggests that there are key differences in several important functional traits, resulting in the ecological niche differences along major dimensions, such as nutrients, light, and temperature (Fig. 4A,B). The regions of the trait space typical for major taxonomic groups differ, both in the freshwater and marine taxa, especially when you consider multiple traits simultaneously (Fig. 4).

In addition to traits that determine the responses of different taxa to abiotic environment, traits describing the relationships with other organisms, such as grazers, parasites, or mutualists, affect fitness and need to be compared across taxa (Litchman et al. 2021; Lürling 2021). A general trend is that HAB-forming taxa often have traits that make them resistant to grazing, through large cell size, coloniality, resting stage formation, and toxicity (Pančić and Kiørboe 2018; Lürling 2021). Viruses seem to readily infect marine and freshwater HAB species (John et al. 2002; Baudoux and Brussaard 2005; Morimoto et al. 2019; Pound and Wilhelm 2021), but tend to be more taxon-specific than grazers. It is possible that in a multidimensional (>3) trait space, that include traits for grazing or viral resistance (e.g., cell size, coloniality, toxicity), the differences between HAB and non-HAB taxa may be even more pronounced.

A better characterization of trait differences between HAB and non-HAB taxa in a *multitrait* space should help determine environmental conditions that favor HAB taxa (Fig. 3). The multitrait information should also help identify the trade-offs among traits, which define different strategies of HAB taxa and could be used to reduce the dimensionality of trait-based models (Litchman et al. 2015*a*; Ehrlich et al. 2020).

Trait differences between HAB- and non-HAB taxa are also relevant for detecting HABs remotely and understanding the dynamics of HABs. For example, not only chlorophyll fluorescence, but phycocyanin (pigment present in many cyanobacteria) and other pigment fluorescence is being actively used to estimate cyanobacterial abundance in water bodies (Simis et al. 2005; Wang et al. 2016). Monitoring HAB toxins or genes associated with toxin production (e.g., domoic acid synthesis associated genes *dab*, Brunson et al. 2018) is also informative.

I discussed how traits differ across the main taxonomic groups, some of which contain HAB-forming species. A related but largely unexplored question is whether HAB-producing species differ significantly in their traits from other species *within* the same taxonomic group that are not known to produce blooms. In cyanobacteria, just a handful of species and genera (e.g., *Microcystis aeruginosa*, *Dolichospermum flos-aquae*, *Nodularia spumigena*, etc.) are responsible for most of the HAB events globally. Many other cyanobacteria (e.g., *Chroococcus*, *Synechococcus*, *Coelosphaerium*, etc.) are not known to form toxic/harmful blooms. In marine environments, among dinoflagellates and diatoms, only some species are known to produce HABs (genera *Alexandrium*, *Karenia*, *Pseudo-nitzschia*, etc.). A comparison of the traits between HAB- and non-HAB taxa within the taxonomic groups may help explain why only some of the species produce blooms. HAB-forming taxa may have larger ranges of key traits which may make them better adapted to different conditions (Brandenburg et al. 2018). Many studies focus more on HAB taxa, so there is insufficient knowledge of trait values of the non HAB-producing taxa.

Trait differences across HAB taxa

Although HAB-forming taxonomic groups appear to have trait values that differ from the non-HAB taxa, they are not monolithic groups (Fig. 3). There is a considerable variation in key traits *among* HAB species within the taxonomic groups. Characterizing these trait differences is important for our understanding of why and when different species bloom. For example, several species/genera form most freshwater cyanobacterial HABs, with probably the most common of them being *Dolichospermum* (*Anabaena*) spp., *Aphanizomenon flos-aquae* and *M. aeruginosa*, sometimes called "Annie, Fannie and Mike". In Lake Mendota, Wisconsin, these three species bloom in different times of year. *Aphanizomenon* and *Dolichospermum* bloom earlier than *Microcystis* (Brock 1985). This may be due to differences in temperature optima among species (Konopka and Brock 1978). There is, however, significant interannual variability in bloom patterns where either a single HAB species dominates for most of the summer or the bloom does not develop at all (Brock 1985). This variability in the dominance patterns may be due to the interactions of various environmental drivers with the corresponding traits of different taxa.

The Laurentian Great Lakes also differ in the identity of their dominant HAB taxa: Lake Erie often experiences massive blooms of *M. aeruginosa* (Steffen et al. 2017), while in Lake Superior, *Dolichospermum* is the dominant HAB genus so far (Sterner et al. 2020). These differences may be associated with the differences in temperature, nutrients (L. Superior is colder and oligotrophic) or some other factors that currently are unidentified. More information on traits of these species and lake-specific strains should help elucidate the reasons. Even within a lake, HABs may consist of different species across space. In Sandusky Bay of Lake Erie, the blooms often consist of *Planktothrix agardhii* and not *Microcystis* (Hampel et al. 2019).

Other common bloom-forming freshwater cyanobacteria include *Planktothrix* spp., *Raphidiopsis* sp., and sometimes *Gloeotrichia echinulata*. Some of these genera/species are nitrogen-fixers, and the ability to fix nitrogen (N) is an important trait that can lead to different effects of environmental factors, nitrogen concentration in particular, on species abundance.

The ability to fix nitrogen is a key trait that can help understand what species can form blooms under different conditions. The seminal work by Val Smith highlighted the

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

importance of N fixation in freshwater cyanobacterial HABs and used this trait to predict blooms, suggesting that low N : P ratios should lead to N-fixer blooms (Smith 1983). However, the blooms of N-fixers were also documented in lakes with high N : P ratios (Noges et al. 2008). Other factors, besides the N : P ratios, may play a role in determining N-fixer dominance. For example, light levels interact with N : P ratios to determine the N-fixer abundance: low N : P ratios stimulated N-fixers only at high light (De Tezanos Pinto and Litchman 2010). High light also stimulated blooms of the N-fixing *N. spumigena* in the Baltic Sea, and decreased light availability led to bloom termination (Wasmund 1997). Therefore, including other traits, such as light-related traits, susceptibility to grazing or temperature-related traits would allow a more nuanced view of the N-fixer blooms and could improve our predictions.

It is likely that differences in eco-physiological traits among different HAB-forming genera and species lead to their dominance under different conditions, both at present and in the future. For example, one possible explanation for the dominance of *Planktothrix* in Sandusky Bay of Lake Erie is its lower half-saturation constant for ammonium uptake and better competitive abilities for ammonium (Hampel et al. 2019). A study of cyanobacterial abundance in Swedish lakes showed that different environmental drivers, such as dissolved organic matter (DOM), pH, and N : P ratios stimulated the growth of different cyanobacterial genera, for example, increasing DOM stimulated *Dolichospermum* but suppressed *Chroococcus* (Freeman et al. 2020), likely due to their differences in traits.

Significant interspecific trait differences have also been reported across the marine HAB taxa. Within the genus *Alexandrium*, there is a significant variation in toxicity, with broad latitudinal trends (decreasing toxicity from north to south in the North Atlantic, Anderson et al. 1994). Three genera of HAB-producing dinoflagellates from the Gulf of Maine responded differently to increasing temperature and acidification: the growth rate of *Alexandrium catenella* decreased, while *Scrippsiella* sp. and *Amphidinium carerae*'s growth rates increased under the projected warming and lower pH, suggesting that the taxonomic composition of HABs may shift in the future (Seto et al. 2019).

Therefore, a better knowledge of the relevant traits and their comparison across taxa in a multitrait space can help explain or even predict what HAB taxa would produce blooms. Comparing the traits of different bloom-forming taxa, as well as contrasting them with the nonbloom-forming species may help answer a general ecological question of why some species have the ability to reach extremely high densities and dominate communities.

Intraspecific trait differences in different HAB species

Although it is clear that common species and genera of HAB phytoplankton differ in their traits, there are also significant differences in traits *within* the HAB species (Schaeffer et al. 2007; Thomas and Litchman 2016; Willis et al. 2016; Dick et al. 2021), which is similar to other groups of phytoplankton and other organisms (Rynearson and Armbrust 2005; Violle et al. 2012). Characterizing the intraspecific variation in traits is important for several reasons (Fig. 3). First, the eco-physiological trait diversity within a species is often due to the presence of multiple genotypes in a population, and this genotypic and phenotypic diversity allows species to potentially adapt to changing environmental conditions, including the anthropogenic global change. The genotypes with higher fitness under novel conditions would be selected for and the population would experience evolutionary rescue (Bell and Gonzalez 2009; Klausmeier et al. 2020*b*). The resulting trait values in a population would change compared to the initial values, thus leading to evolutionary adaptation. Determining the degree of trait variation within a population or a species can thus help assess the evolutionary potential of HAB taxa. For example, a comparison of thermal curves in *Microcystis* showed that although many strains isolated from Michigan lakes had temperature optima for growth below 30°C, there was also a strain that had a higher T_{opt}, suggesting that there is relevant genotypic and phenotypic diversity that may allow this species to maintain high growth rates under warming conditions (Thomas and Litchman 2016).

Second, assessing the intraspecific variation in traits will also inform us about the width of the environmental niches of HAB taxa, as environmental tolerances may widen if there are genotypes with different trait values; for example, considerable intraspecific variation in T_{opt} could allow species to persist under a wide range of temperatures (Fig. 2, Thomas and Litchman 2016). A survey of photosynthetic traits in multiple strains of the toxic dinoflagellate *Karenia brevis* revealed low light- and high light-adapted strategies, suggesting that the species can survive in different light conditions (Schaeffer et al. 2007).

At the same time, characterizing and comparing intraspecific trait variation in different environments allows us to determine the degree of local adaptation (Kawecki and Ebert 2004). There are few studies exploring local trait adaptation in HAB taxa, but the existing evidence suggests that it may be common in both marine and freshwater environments. The toxic dinoflagellate *Alexandrium ostenfeldii* showed a significant genetic differentiation among different geographic locations, suggesting local adaptation (Tahvanainen et al. 2012; Brandenburg et al. 2021). Similarly, the freshwater cyanobacterium *Microcystis* showed genetic divergence across lakes in Michigan (Wilson et al. 2005). An invasive cyanobacterium *R.* (*C.*) *raciborskii* may be adapted to local temperatures, as the strain from the Midwest had lower T_{opt} than the Florida strains (Fig. 2) (Thomas and Litchman 2016). Studies that investigate the geographic patterns in both genetic and trait diversity would be especially illuminating.

A better characterization of the traits of different genotypes and trade-offs between traits within a species or a population can help predict possible changes in toxicity. Strains of

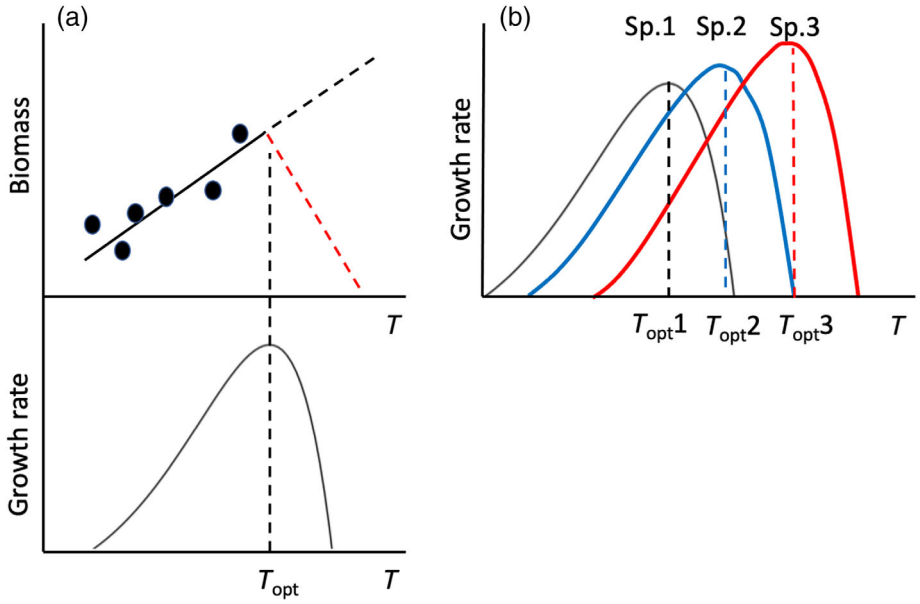


Fig. 5. Simplified examples of how including traits into the predictions of HABs may be useful. (**a**) Upper panel shows a statistical linear model where the biomass of cyanobacteria increases with temperature (solid black line). In the absence of knowledge of the temperature traits (lower panel), for the temperatures increasing further, the model would predict biomass increase (black dashed line on the upper panel). However, including the temperature trait information may lead to a change in the relationship (red dashed line) and predict a decrease in biomass because of the decline in growth rate of that cyanobacterium with increasing temperature. (**b**) The common focus on the individual HAB species (e.g., Sp. 1) may erroneously predict a decline in biomass when temperature increases above the T_{opt} for Sp. 1. However, depending on what species are present, the cyanobacterial biomass may still increase with increasing temperature or decline. The temperature where the decline may happen will depend on the temperature traits of the HAB species in the community.

Microcystis exhibited a trade-off between competitive ability for light and toxicity (Kardinaal et al. 2007). Consequently, with the progression of the *Microcystis* bloom and decreasing light availability, less toxic strains with greater light competitive ability replaced more toxic strains that were poor light competitors, thus decreasing the bloom toxicity (Kardinaal et al. 2007). The knowledge of the trait variation for multiple strains/genotypes within a species may help assess the outcomes of competitive interactions between different HAB genera, as the outcome of competition between *Microcystis* and *Cylindrospermopsis* depended on what strains were involved (Xiao et al. 2017).

Intraspecific trait variation arises either due to trait differences across genotypes or due to physiological plasticity (Violle et al. 2012). Many studies have documented plasticity of many traits in marine and freshwater HAB taxa, such as CO_2 uptake, degree of coloniality, inducible defenses, toxicity, nitrogen fixation, and so on (De Tezanos Pinto and Litchman 2010; Wohlrab et al. 2017; Ji et al. 2020; Lürling 2021). Different genotypes may have different degrees of phenotypic plasticity, as in a toxic dinoflagellate *A. ostenfeldii* (Brandenburg et al. 2021). Comparing the degree of phenotypic plasticity within (and across) genotypes to the trait variation across genotypes would help determine if the fitness under future conditions would be influenced more by evolutionary adaptation or by acclimation.

Information on the degree of intraspecific trait variation in HAB taxa is crucial for explaining the ecological success of major toxic bloom-forming species in marine and freshwater environments. Studies on key HAB taxa, such as the marine dinoflagellate *Alexandrium* spp., freshwater cyanobacterium *Microcystis* spp., and others show considerable genotypic and phenotypic diversity, which may be one of the reasons for their global distribution and frequent blooms (Anderson et al. 2012; Brandenburg et al. 2018; Dick et al. 2021). Interestingly, the degree of intraspecific trait variation can differ across populations from different geographic locations, such as in the toxic marine dinoflagellate *A. ostenfeldii* isolated from the Baltic Sea vs. the Netherlands (Brandenburg et al. 2018). It is unknown whether the trait/phenotypic diversity is considerably greater in HAB-producing vs. non-HAB taxa.

Predicting HABs using traits

Surprisingly, using trait-based approaches to predict the dynamics of species and communities in the future is much less developed compared to trait-based explanations of the observed past and present community composition. Incorporating trait-based approaches into predictions should help increase the reliability of predictions and extend their time horizon.

Predictive models for HABs, as for many other ecological phenomena, can be divided into statistical (data-driven) and

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

process-based (mechanistic) models (Ralston and Moore 2020). Statistical models may work reasonably well, especially for the shorter time scale predictions, but may fail on longer time scales, or when the conditions change rapidly and create novel combinations or ranges of environmental drivers, which is likely to happen given the anthropogenic global change. Process-based models that include the physiological dependence of growth and mortality of HAB taxa on environmental factors may predict the dynamics of HABs better (Ralston and Moore 2020; Wang et al. 2022). Adequately parameterizing such models is, however, difficult, as is including all the relevant processes without overly complicating the models.

There are at least three ways in which a trait-based framework can be used to improve predictions. First, the relevant traits can be included as predictors in statistical models, in addition to environmental variables (Fig. 5). Some models that predict species abundances and distributions started including traits as predictors, which improved model performance (Regos et al. 2019; Vesk et al. 2021). Edwards et al. (2013*a*) showed that several eco-physiological traits of phytoplankton, namely the affinity for light and a nutrient (nitrate), and the maximum growth rate were significant predictors of the species' abundance changes in response to seasonally changing environmental conditions in a marine ecosystem, explaining up to 80% of the variance in abundance. Statistical models that include both the environmental factors and functional traits as predictors may work better than the models that only include environmental variables. The multilevel (mixed, hierarchical) statistical models such as the one used by Edwards et al. (2013*a*) and others (Thorson and Minto 2014) can include variation between species due to environmental factors, variation between species not explained by environmental factors and variation not explained by species identity or environmental factors.

Second, empirical trait information can be used to systematically parameterize process-based, mechanistic models of HAB dynamics. Choosing the parameter values and their ranges based on multiple studies helps increase the reliability of parameterizations and guide sensitivity analyses where different parameter values are explored. We have previously used an extensive trait compilation to parameterize a dynamic model of the phytoplankton community composition in the present and future ocean and attempted to identify traits that were the most influential for the predictions (Litchman et al. 2006). Similar approaches can be used to parameterize the phytoplankton community models that include HAB taxa and to determine what traits are the most influential for predicting HAB dynamics under different conditions and thus require more thorough measurements.

Third, we can use trait-based models to augment the existing dynamical (e.g., differential equations-based models of population density dynamics) or statistical models, which should improve our predictions of the blooms on longer time scales and in novel conditions, and would allow the possibility for different HAB taxa to become dominant. Strictly speaking, trait-based models are models that predict what traits and trait combinations would be selected for under different conditions and are, thus, species/taxon agnostic (Falster et al. 2017; Klausmeier et al. 2020*a*). Therefore, contrary to the popular misconception, the models that include traits of species, as many dynamical models that have species or functional groups do, are taxon-based, not trait-based. Such commonly used species or functional group models are useful as well, especially when parameterized based on reliable empirical trait data and when they include relevant mechanisms, such as resource-dependent growth, mortality from grazing and other sources, and the effects of environmental drivers (temperature, pH, physical mixing, etc.). However, they may not adequately predict the rise of novel HAB taxa not included in the models.

Combining different types of models may be advantageous and provide better predictions of HABs/water quality under changing conditions. Such hybrid approaches may include agent-based modeling of the dynamics of toxin genes and more standard hydrodynamic models (Hellweger et al. 2022) or statistical models of oxygen concentration with varying coefficients (not the HAB dynamics per se) and a hydrodynamic model (Deyle et al. 2022). Linking trait-based models or species-based models parameterized using extensive empirical trait data with hydrodynamic models forced by the present or future climate scenarios may provide a robust way to predict when and what HAB taxa would proliferate.

Trait-based models can inform us what trait values would lead to high fitness (often approximated by growth rate) in different environments (Litchman and Klausmeier 2008). For the models that focus on HABs, relevant traits include traits that determine the growth dependence on environmental factors, both abiotic and biotic (e.g., grazing or parasitism) (Table 1). The models could predict trait value combinations in a multidimensional trait space that lead to the highest fitness under given conditions. Consequently, the taxa (groups, species, or strains) that have the combinations of traits closest to the predicted combinations would likely be selected for under given environmental conditions. Then matching the predicted trait combinations with existing trait data for different species could help identify what species would likely dominate (Fig. 4).

Mechanistic trait-based models may predict a particular combination of traits that results in the highest fitness under given environmental conditions. It is possible that some range of trait combinations may have similar fitness, leading to the possibility of several species co-occurring, especially if species with lower fitness are supported by immigration or changing selection regimes (Koffel et al. 2022). Theoretical models may also predict more than one optimal trait value or trait combination (Litchman et al. 2009; Kremer and Klausmeier 2017), which may make coexistence of different

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ecological strategies, possibly represented by different taxonomic groups, likely.

One relevant trait-based modeling approach that was used successfully to explain different community composition of phytoplankton in the world ocean was to "seed" a community with species/phenotypes that have different trait combinations (e.g., from low- to high-light adapted, from low to high maximum growth rates, T_{opt}, etc.) and simulate community dynamics under diverse environmental conditions to determine what trait combinations win under given conditions (Follows et al. 2007). Those predicted trait values can then be matched with the empirical trait distributions for different phytoplankton, including HAB taxa to determine the probability of a bloom.

The advantage of such models that explore all the possible trait space and determine trait combinations associated with high fitness under different environmental conditions is that they are not constrained by what species/taxa they model and, thus, allow new species to become dominant. For example, a model that only includes *Microcystis* sp. would not be able to predict the development of a *Planktothrix* or *Dolichospermum* bloom if the conditions change, while trait-based models could. Trait-based models should be especially useful for predicting HABs under rapidly changing conditions or on longer time scales, when an expansion of different species may be more likely.

Future directions

Combining the data-driven and modeling trait-based approaches to understand and predict HABs in freshwater and marine environments is a promising area of research. There are some key directions that should speed up progress. First, we need more information on trait values of different HAB taxa to compare them systematically with the traits of non-HAB taxa and to parameterize mechanistic models. The traits that describe the growth dependence of HAB taxa on environmental (abiotic and biotic) drivers (Table 1) are especially relevant. In addition to traditional ways of measuring traits (laboratory experiments), we need to develop new approaches, such as inferring traits that can be used in predictive models from the 'omic (genomic, transcriptomic, proteomic, and metabolomic) data (Hennon and Dyhrman 2020; Strzepek et al. 2022) or the environmental distribution data using machine learning or similar techniques (Newman and Furbank 2021). Using genomic information to predict the toxic potential of the blooms is widespread (Janse et al. 2004; Brunson et al. 2018; Panksep et al. 2020), but inferring other traits such as growth rates or nutrient uptake is much more complicated. The 'omic approaches are being increasingly used to characterize the responses of different HAB species and strains to environmental conditions, which may aid HAB predictions (Hennon and Dyhrman 2020). Combining 'omics data with trait information for many species may allow identifying the 'omic signatures for some traits and help derive the missing traits. In parallel, species responses to the environment and the associated traits can also be derived from the environmental time series using machine learning or similar computational approaches, but such potential remains underexplored (Thomas et al. unpublished).

Compared to the traits related to abiotic drivers, traits that characterize biotic interactions, such as predation, parasitism or mutualism, are less frequently characterized (Litchman et al. 2021), and should be one of the research foci in HAB ecology. For example, little is known about what traits may protect from viral attacks or other parasites, and if there are trade-offs between the defense and other traits, such as nutrient uptake capabilities (Menge and Weitz 2009; Litchman et al. 2021).

Characterizing the multitrait surfaces for HAB and non-HAB taxa would allow us to understand whether the given environmental conditions (combinations of nutrients, temperature, light, water mixing, etc.) would lead to blooms of harmful algae. At the same time, the knowledge of traits within HAB taxa, across different species, would allow us to determine what species would be most likely to bloom. Moreover, better characterizing the *intraspecific* variation in traits is essential for predicting the ability of different HAB taxa to adapt to changing environmental conditions, including the anthropogenic global change. All this trait information should be compiled into a HAB trait database, similar to the trait databases for other organisms, such as terrestrial plants and fish (Angermeier and Frimpong 2011; Kattge et al. 2020). Such databases enabled numerous fundamental advances in community ecology and would be essential for HAB research.

Although at least some trait data for key planktonic HAB taxa exist, trait information on the benthic HABs is lacking. Benthic toxic cyanobacteria proliferate in many water bodies, especially running waters, but the measurements of their traits are scant (Wood et al. 2020). Future trait work should increase focus on benthic forms. In addition, many planktonic HAB taxa can be found in benthic habitats, for example, as resting stages (Ståhl-Delbanco et al. 2003; Ellegaard and Ribeiro 2018). Characterizing HAB traits during the benthic stage is necessary to understand different parts of the life cycle of HAB taxa.

Another direction is developing mechanistic models that include key traits and can predict trait selection under different future scenarios (mixing, nutrient levels, temperature, etc.). Such models should include not only the abiotic but biotic controls as well, such as grazing, parasitism, and mutualism. Parameterizing these models using extensive trait databases will increase model realism. The models should then be tested with different types of empirical data, with different temporal (e.g., from weekly to multiyear) and spatial (individual lakes vs. regional or global) resolution.

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References

Anderson, D. M., D. M. Kulis, G. J. Doucette, J. C. Gallagher, and E. Balech. 1994. Biogeography of toxic dinoflagellates in the genus *Alexandrium* from the northeastern United States and Canada. Marine Biology **120**: 467–478. doi:10.1007/BF00680222

Anderson, D. M., A. D. Cembella, and G. M. Hallegraeff [eds.]. 1998, *The physiological ecology of harmful algal blooms*. Springer-Verlag.

Anderson, D. M., P. Hoagland, Y. Kaoru, and A. W. White. 2000, *Estimated annual economic impacts of harmful algal blooms (HABs) in the United States*. NOAA.

Anderson, D. M., T. J. Alpermann, A. D. Cembella, Y. Collos, E. Masseret, and M. Montresor. 2012. The globally distributed genus *Alexandrium*: Multifaceted roles in marine ecosystems and impacts on human health. Harmful Algae **14**: 10–35. doi:10.1016/j.hal.2011.10.012

Angermeier, P. L., and E. A. Frimpong. 2011. FishTraits: A database of ecological and life-history traits of freshwater fishes of the United States. Fisheries **34**: 487–495.

Annunziata, R., and others. 2022. Trade-off between sex and growth in diatoms: Molecular mechanisms and demographic implications. Sci. Adv. **8**: eabj9466.

Baudoux, A. C., and C. P. D. Brussaard. 2005. Characterization of different viruses infecting the marine harmful algal bloom species *Phaeocystis globosa*. Virology **341**: 80–90. doi:10.1016/j.virol.2005.07.002

Bell, G., and A. Gonzalez. 2009. Evolutionary rescue can prevent extinction following environmental change. Ecol. Lett. **12**: 942–948. doi:10.1111/j.1461-0248.2009.01350.x

Brandenburg, K. M., S. Wohlrab, U. John, A. Kremp, J. Jerney, B. Krock, and D. B. Van de Waal. 2018. Intraspecific trait variation and trade-offs within and across populations of a toxic dinoflagellate. Ecol. Lett. **21**: 1561–1571. doi:10.1111/ele.13138

Brandenburg, K. M., B. Krock, H. C. L. Klip, A. Sluijs, P. Garbeva, and D. B. Van de Waal. 2021. Intraspecific variation in multiple trait responses of *Alexandrium ostenfeldii* towards elevated pCO_2. Harmful Algae **101**: 101970. doi:10.1016/j.hal.2020.101970

Brock, T. D. 1985, *A eutrophic lake: Lake Mendota, Wisconsin*. Springer-Verlag.

Bruggeman, J. 2011. A phylogenetic approach to the estimation of phytoplankton traits. J. Phycol. **47**: 52–65. doi:10.1111/j.1529-8817.2010.00946.x

Brunson, J. K., S. M. K. McKinnie, J. R. Chekan, J. P. McCrow, Z. D. Miles, E. M. Bertrand, V. A. Bielinski, H. Luhavaya, M. Oborník, G. J. Smith, D. A. Hutchins, A. E. Allen, and B. S. Moore. 2018. Biosynthesis of the neurotoxin domoic acid in a bloom-forming diatom. Science **361**: 1356–1358. doi:10.1126/science.aau0382

Butler, E. E., and others. 2017. Mapping local and global variability in plant trait distributions. Proc. Nat. Acad. Sci. **114**: E10937–E10946.

Chorus, I., and M. Welker [eds.]. 2021, *Toxic cyanobacteria in water*, 2nd Edition. World Health Organization. doi:10.3390/toxins14010004

Chuckran, P. F., B. A. Hungate, E. Schwartz, and P. Dijkstra. 2021. Variation in genomic traits of microbial communities among ecosystems. FEMS Microb. **2**: xtab020.

De Tezanos Pinto, P., and E. Litchman. 2010. The interactive effects of N:P ratios and light on nitrogen-fixer abundance. Oikos **119**: 567–575. doi:10.1111/j.1600-0706.2009.17924.x

Deyle, E. R., D. Bouffard, V. Frossard, R. Schwefel, J. Melack, and G. Sugihara. 2022. A hybrid empirical and parametric approach for managing ecosystem complexity: Water quality in Lake Geneva under nonstationary futures. Proc. Nat. Acad. Sci. **119**: e2102466119.

Diaz, J. M., and S. Plummer. 2018. Production of extracellular reactive oxygen species by phytoplankton: Past and future directions. J. Plankton Res. **40**: 655–666. doi:10.1093/plankt/fby039

Díaz, S., and others. 2016. The global spectrum of plant form and function. Nature **529**: 167–171. doi:10.1038/nature16489

Dick, G. J., and others. 2021. The genetic and ecophysiological diversity of *Microcystis*. Environ. Microbiol. **23**: 7278–7313. doi:10.1111/1462-2920.15615

Edwards, K. F., C. A. Klausmeier, and E. Litchman. 2011. Evidence for a three-way tradeoff between nitrogen and phosphorus competitive abilities and cell size in phytoplankton. Ecology **92**: 2085–2095. doi:10.1890/11-0395.1

Edwards, K. F., M. K. Thomas, C. A. Klausmeier, and E. Litchman. 2012. Allometric scaling and taxonomic variation in nutrient utilization traits and growth rates of marine and freshwater phytoplankton. Limnol. Oceanogr. **57**: 554–566. doi:10.4319/lo.2012.57.2.0554

Edwards, K. F., E. Litchman, and C. A. Klausmeier. 2013*a*. Functional traits explain phytoplankton community structure and seasonal dynamics in a marine ecosystem. Ecol. Lett. **16**: 56–63.

Edwards, K. F., E. Litchman, and C. A. Klausmeier. 2013*b*. Functional traits explain phytoplankton responses to environmental gradients across lakes of the United States. Ecology **94**: 1626–1635.

Edwards, K. F., M. K. Thomas, C. A. Klausmeier, and E. Litchman. 2015. Light and phytoplankton growth: Allometry, taxonomic variation, and biogeography. Limnol. Oceanogr. **60**: 540–552. doi:10.1002/lno.10033

Edwards, K. F., M. K. Thomas, C. A. Klausmeier, and E. Litchman. 2016. Phytoplankton growth and the interaction of light and temperature: A synthesis at the species and community level. Limnol. Oceanogr. **61**: 1232–1244. doi:10.1002/lno.10282

Edwards, K. F., and G. F. Steward. 2018. Host traits drive viral life histories across phytoplankton viruses. Am. Nat. **191**: 566–581. doi:10.1086/696849

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ehrlich, E., N. J. Kath, and U. Gaedke. 2020. The shape of a defense-growth trade-off governs seasonal trait dynamics in natural phytoplankton. ISME J. **14**: 1451–1462. doi:10.1038/s41396-020-0619-1

Ellegaard, M., and S. Ribeiro. 2018. The long-term persistence of phytoplankton resting stages in aquatic "seed banks". Biol. Rev. **93**: 166–183. doi:10.1111/brv.12338

Facey, J. A., S. C. Apte, and S. M. Mitrovic. 2019. A review of the effect of trace metals on freshwater Cyanobacterial growth and toxin production. Toxins (Basel) **11**: 643. doi:10.3390/toxins11110643

Falster, D. S., A. Brannstrom, M. Westoby, and U. Dieckmann. 2017. Multitrait successional forest dynamics enable diverse competitive coexistence. Proc. Nat. Acad. Sci. USA **114**: E2719–E2728.

Figueroa, R. I., C. Dapena, I. Bravo, and A. Cuadrado. 2015. The hidden sexuality of *Alexandrium minutum*: An example of overlooked sex in dinoflagellates. PLoS One **10**: e0142667.

Fogg, G. E., and A. E. Walsby. 1971. Buoyancy regulation and the growth of planktonic blue-green algae. Mitt. Intrnat. Verein. Limnol. **19**: 182–188.

Follows, M. J., S. Dutkiewicz, S. Grant, and S. W. Chisholm. 2007. Emergent biogeography of microbial communities in a model ocean. Science **315**: 1843–1846. doi:10.1126/science.1138544

Freeman, E. C., I. F. Creed, B. Jones, and A.-K. Bergström. 2020. Global changes may be promoting a rise in select cyanobacteria in nutrient-poor northern lakes. Global Change Biol. **26**: 4966–4987. doi:10.1111/gcb.15189

Funk, J. L., and others. 2017. Revisiting the Holy Grail: Using plant functional traits to understand ecological processes. Biol. Rev. **92**: 1156–1173. doi:10.1111/brv.12275

Garnier, E., M.-L. Navas, and K. Grigulis. 2016, *Plant functional diversity: Organism traits, community structure, and ecosystem properties*. Oxford Univ. Press.

Grover, J. P. 1997, *Resource competition*. Chapman and Hall.

Hallegraeff, G. M., and others. 2021. Perceived global increase in algal blooms is attributable to intensified monitoring and emerging bloom impacts. Commun. Earth Environ. **2**: 117.

Hampel, J. J., M. J. McCarthy, M. Neudeck, G. S. Bullerjahn, R. M. L. McKay, and S. E. Newell. 2019. Ammonium recycling supports toxic *Planktothrix* blooms in Sandusky Bay, Lake Erie: Evidence from stable isotope and metatranscriptome data. Harmful Algae **81**: 42–52. doi:10.1016/j.hal.2018.11.011

Heisler, J., and others. 2008. Eutrophication and harmful algal blooms: A scientific consensus. Harmful Algae **8**: 3–13. doi:10.1016/j.hal.2008.08.006

Hellweger, F. L., R. M. Martin, F. Eigemann, D. J. Smith, G. J. Dick, and S. W. Wilhelm. 2022. Models predict planned phosphorus load reduction will make Lake Erie more toxic. Science **376**: 1001–1005. doi:10.1126/science.abm6791

Hennon, G. M. M., and S. T. Dyhrman. 2020. Progress and promise of omics for predicting the impacts of climate change on harmful algal blooms. Harmful Algae **91**: 101587. doi:10.1016/j.hal.2019.03.005

Herrero, A., and E. Flores. 2008, *The cyanobacteria: Molecular biology, genomics and evolution*. Caister Academic Press.

Ho, J. C., A. M. Michalak, and N. Pahlevan. 2019. Widespread global increase in intense lake phytoplankton blooms since the 1980s. Nature **574**: 667–670. doi:10.1038/s41586-019-1648-7

Hong, H.-H., and others. 2016. The exceptionally large genome of the harmful red tide dinoflagellate *Cochlodinium polykrikoides* Margalef (Dinophyceae): determination by flow cytometry. Algae **31**: 373–378.

Hoke, A. K., and others. 2021. Genomic signatures of Lake Erie bacteria suggest interaction in the *Microcystis* phycosphere. PLoS One **16**: e0257017. doi:10.1371/journal.pone.0257017

Huisman, J., G. A. Codd, H. W. Paerl, B. W. Ibelings, J. M. H. Verspagen, and P. M. Visser. 2018. Cyanobacterial blooms. Nat. Rev. Microbiol. **16**: 471–483. doi:10.1038/s41579-018-0040-1

Janse, I., W. E. Kardinaal, M. Meima, J. Fastner, P. M. Visser, and G. Zwart. 2004. Toxic and nontoxic *Microcystis* colonies in natural populations can be differentiated on the basis of rRNA gene internal transcribed spacer diversity. Appl. Environ. Microbiol. **70**: 3979–3987. doi:10.1128/AEM.70.7.3979-3987.2004

Ji, X., J. M. H. Verspagen, D. B. Van de Waal, B. Rost, and J. Huisman. 2020. Phenotypic plasticity of carbon fixation stimulates cyanobacterial blooms at elevated CO_2. Sci. Adv. **6**: eaax2926.

John, H. P., H. Lee, G. Dale, S. Terri, G. Mike, R. Bill, and S. Karen. 2002. A filterable lytic agent obtained from a red tide bloom that caused lysis of *Karenia brevis* (*Gymnodinum breve*) cultures. Aquat. Microb. Ecol. **27**: 21–27.

Kardinaal, W. E. A., L. Tonk, I. Janse, S. Hol, P. Slot, J. Huisman, and P. M. Visser. 2007. Competition for light between toxic and nontoxic strains of the harmful cyanobacterium *Microcystis*. Appl. Environ. Microbiol. **73**: 2939–2946. doi:10.1128/AEM.02892-06

Kattge, J., and others. 2020. TRY plant trait database—Enhanced coverage and open access. Global Change Biol. **26**: 119–188. doi:10.1111/gcb.14904

Kawecki, T. J., and D. Ebert. 2004. Conceptual issues in local adaptation. Ecol. Lett. **7**: 1225–1241. doi:10.1111/j.1461-0248.2004.00684.x

Klausmeier, C. A., C. T. Kremer, and T. Koffel. 2020*a*. Trait-based ecological and eco-evolutionary theory, p. 161–194. *In* K. McCann and G. Gellner [eds.], *Theoretical ecology: Concepts and applications*. Oxford Univ. Press.

Klausmeier, C. A., M. M. Osmond, C. T. Kremer, and E. Litchman. 2020*b*. Ecological limits to evolutionary rescue. Philos. Trans. R. Soc. B Biol. Sci. **375**: 20190453.

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Klemer, A. R. 1983. Cyanobacterial buoyancy regulation and blooms, *In Lake restoration, protection and management*. p. 234–236. North American Lake Management Society.

Koffel, T., K. Umemura, E. Litchman, and C. A. Klausmeier. 2022. A general framework for species-abundance distributions: Linking traits and dispersal to explain commonness and rarity. Ecol. Lett. **25**: 2359–2371. doi:10.1111/ele.14094

Kokocinski, M., K. Stefaniak, J. Mankiewicz-Boczek, K. Izydorczyk, and J. Soininen. 2010. The ecology of the invasive cyanobacterium *Cylindrospermopsis raciborskii* (Nostocales, Cyanophyta) in two hypereutrophic lakes dominated by *Planktothrix agardhii* (Oscillatoriales, Cyanophyta). Eur. J. Phycol. **45**: 365–374. doi:10.1080/09670262.2010.492916

Konopka, A., and T. D. Brock. 1978. Effect of temperature on blue-green algae (cyanobacteria) in sake Mendota. Appl. Environ. Microbiol. **36**: 572–576. doi:10.1128/aem.36.4.572-576.1978

Kremer, C. T., and C. A. Klausmeier. 2017. Species packing in eco-evolutionary models of seasonally fluctuating environments. Ecol. Lett. **20**: 1158–1168. doi:10.1111/ele.12813

Kruk, C., V. L. M. Huszar, E. Peeters, S. Bonilla, L. Costa, M. Lurling, C. S. Reynolds, and M. Scheffer. 2010. A morphological classification capturing functional variation in phytoplankton. Freshw. Biol. **55**: 614–627. doi:10.1111/j.1365-2427.2009.02298.x

Kruk, C., and others. 2017. A multilevel trait-based approach to the ecological performance of *Microcystis aeruginosa* complex from headwaters to the ocean. Harmful Algae **70**: 23–36. doi:10.1016/j.hal.2017.10.004

Kruk, C., M. Devercelli, and V. L. Huszar. 2021. Reynolds functional groups: A trait-based pathway from patterns to predictions. Hydrobiologia **848**: 113–129. doi:10.1007/s10750-020-04340-9

Kudela, R. M., S. Seeyave, and W. P. Cochlan. 2010. The role of nutrients in regulation and promotion of harmful algal blooms in upwelling systems. Prog. Oceanogr. **85**: 122–135.

Lamanna, C., and others. 2014. Functional trait space and the latitudinal diversity gradient. Proc. Nat. Acad. Sci. USA **111**: 13745–13750.

Larsson, J., J. A. A. Nylander, and B. Bergman. 2011. Genome fluctuations in cyanobacteria reflect evolutionary, developmental and adaptive traits. BMC Evol. Biol. **11**: 187.

Lavorel, S., and E. Garnier. 2002. Predicting changes in community composition and ecosystem functioning from plant traits: Revisiting the Holy Grail. Funct. Ecol. **16**: 545–556. doi:10.1046/j.1365-2435.2002.00664.x

Lewis, N. I., W. Xu, S. K. Jericho, H. J. Kreuzer, M. H. Jericho, and A. D. Cembella. 2006. Swimming speed of three species of *Alexandrium* (Dinophyceae) as determined by digital in-line holography. Phycologia **45**: 61–70.

Li, R., S. Zhu, J. Lian, H. Zhang, H. Liu, W. Ye, and Q. Ye. 2021. Functional traits are good predictors of tree species abundance across 101 subtropical forest species in China. Front. Plant Sci. **12**: 41577. doi:10.3389/fpls.2021.763618

Litchman, E., C. A. Klausmeier, J. R. Miller, O. M. Schofield, and P. G. Falkowski. 2006. Multi-nutrient, multi-group model of present and future oceanic phytoplankton communities. Biogeosciences **3**: 585–606.

Litchman, E., C. A. Klausmeier, O. M. Schofield, and P. G. Falkowski. 2007. The role of functional traits and trade-offs in structuring phytoplankton communities: Scaling from cellular to ecosystem level. Ecol. Lett. **10**: 1170–1181. doi:10.1111/j.1461-0248.2007.01117.x

Litchman, E., and C. A. Klausmeier. 2008. Trait-based community ecology of phytoplankton. Annu. Rev. Ecol. Evol. Syst. **39**: 615–639. doi:10.1146/annurev.ecolsys.39.110707.173549

Litchman, E., C. A. Klausmeier, and K. Yoshiyama. 2009. Contrasting size evolution in marine and freshwater diatoms. Proc. Natl. Acad. Sci. USA **106**: 2665–2670. doi:10.1073/pnas.0810891106

Litchman, E., K. F. Edwards, and C. A. Klausmeier. 2015*a*. Microbial resource utilization traits and trade-offs: Implications for commuity structure, functioning and biogeochemical impacts at present and in the future. Front. Microbiol. **6**: 254.

Litchman, E., P. de Tezanos Pinto, K. F. Edwards, C. A. Klausmeier, C. T. Kremer, and M. K. Thomas. 2015*b*. Global biogeochemical impacts of phytoplankton in the past, present, and future: A trait-based perspective. J. Ecol. **103**: 1384–1396.

Litchman, E., K. F. Edwards, and P. W. Boyd. 2021. Towards trait-based food webs: Universal traits and trait matching in planktonic predator–prey and host–parasite relationships. Limnol. Oceanogr. **66**: 3857–3872. doi:10.1002/lno.11924

Liu, H., Z. C. Zheng, B. Young, and T. D. Harris. 2019. Three-dimensional numerical modeling of the cyanobacterium *Microcystis* transport and its population dynamics in a large freshwater reservoir. Ecol. Model. **398**: 20–34. doi:10.1016/j.ecolmodel.2019.01.022

Lürling, M. 2021. Grazing resistance in phytoplankton. Hydrobiologia **848**: 237–249. doi:10.1007/s10750-020-04370-3

Margalef, R. 1978. Life forms of phytoplankton as survival alternatives in an unstable environment. Oceanol. Acta **1**: 493–509.

McGill, B. J., B. J. Enquist, E. Weiher, and M. Westoby. 2006. Rebuilding community ecology from functional traits. Trends Ecol. Evol. **21**: 178–185. doi:10.1016/j.tree.2006.02.002

McGowan, J. A., E. R. Deyle, H. Ye, M. L. Carter, C. T. Perretti, K. D. Seger, A. de Verneil, and G. Sugihara. 2017. Predicting

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

coastal algal blooms in southern California. Ecology **98**: 1419–1433. doi:10.1002/ecy.1804

Menge, D. N. L., and J. S. Weitz. 2009. Dangerous nutrients: Evolution of phytoplankton resource uptake subject to virus attack. J. Theor. Biol. **257**: 104–115. doi:10.1016/j.jtbi.2008.10.032

Moestrup, Ø., R. Akselmann-Cardella, S. F. C. Churro, M. Hoppenrath, M. Iwataki, J. Larsen, N. Lundholm, and A. Zingone. 2009. IOC-UNESCO taxonomic reference list of harmful micro algae. Available from https://www.marinespecies.org/hab/

Morimoto, D., K. Tominaga, Y. Nishimura, N. Yoshida, S. Kimura, Y. Sako, and T. Yoshida. 2019. Cooccurrence of broad- and narrow-host-range viruses infecting the bloom-forming toxic cyanobacterium *Microcystis aeruginosa*. Appl. Environ. Microbiol. **85**: e01170-01119. doi:10.1128/AEM.01170-19

Mousing, E. A., K. Richardson, and M. Ellegaard. 2018. Global patterns in phytoplankton biomass and community size structure in relation to macronutrients in the open ocean. Limnol. Oceanogr. **63**: 1298–1312. doi:10.1002/lno.10772

Newman, S. J., and R. T. Furbank. 2021. Explainable machine learning models of major crop traits from satellite-monitored continent-wide field trial data. Nat. Plants **7**: 1354–1363. doi:10.1038/s41477-021-01001-0

Noges, T., R. Laugaste, P. Noges, and I. Tonno. 2008. Critical N: P ratio for cyanobacteria and N-2-fixing species in the large shallow temperate lakes Peipsi and Vortsjarv, North-East Europe. Hydrobiologia **599**: 77–86. doi:10.1007/s10750-007-9195-x

Paerl, H. W., and J. Huisman. 2009. Climate change: A catalyst for global expansion of harmful cyanobacterial blooms. Environ. Microbiol. Rep. **1**: 27–37. doi:10.1111/j.1758-2229.2008.00004.x

Pančić, M., and T. Kiørboe. 2018. Phytoplankton defence mechanisms: Traits and trade-offs. Biol. Rev. **93**: 1269–1303. doi:10.1111/brv.12395

Panksep, K., M. Tamm, E. Mantzouki, A. Rantala-Ylinen, R. Laugaste, K. Sivonen, O. Tammeorg, and V. Kisand. 2020. Using microcystin gene copies to determine potentially toxic blooms, example from a shallow eutrophic Lake Peipsi. Toxins **12**: 211. doi:10.3390/toxins12040211

Papoulis, S. E., S. W. Wilhelm, D. Talmy, E. R. Zinser, and M. A. Moran. 2021. Nutrient loading and viral memory drive accumulation of restriction modification systems in bloom-forming Cyanobacteria. Mbio **12**: e00873–e00821.

Penone, C., and others. 2014. Imputation of missing data in life-history trait datasets: Which approach performs the best? Methods in Ecology and Evolution **5**: 961–970. doi:10.1111/2041-210X.12232

Pound, H. L., and S. W. Wilhelm. 2021. Tracing the active genetic diversity of *Microcystis* and *Microcystis* phage through a temporal survey of Taihu. PLoS One **15**: e0244482.

Ralston, D. K., and S. K. Moore. 2020. Modeling harmful algal blooms in a changing climate. Harmful Algae **91**: 101729. doi:10.1016/j.hal.2019.101729

Raven, J. A., C. J. Gobler, and P. J. Hansen. 2020. Dynamic CO_2 and pH levels in coastal, estuarine, and inland waters: Theoretical and observed effects on harmful algal blooms. Harmful Algae **91**: 101594. doi:10.1016/j.hal.2019.03.012

Recknagel, F., T. Zohary, J. Rücker, P. T. Orr, C. C. Branco, and B. Nixdorf. 2019. Causal relationships of *Raphidiopsis* (formerly *Cylindrospermopsis*) dynamics with water temperature and N:P-ratios: A meta-analysis across lakes with different climates based on inferential modelling. Harmful Algae **84**: 222–232. doi:10.1016/j.hal.2019.04.005

Regos, A., L. Gagne, D. Alcaraz-Segura, J. P. Honrado, and J. Domínguez. 2019. Effects of species traits and environmental predictors on performance and transferability of ecological niche models. Sci. Rep. **9**: 4221. doi:10.1038/s41598-019-40766-5

Reynolds, C. S. 1984, *The ecology of freshwater phytoplankton*. Cambridge Univ. Press.

Reynolds, C. S. 1997, *Vegetation processes in the pelagic: A model for ecosystem theory*. Inter-Research.

Reynolds, C. S. 2006, *The ecology of phytoplankton*. Cambridge Univ. Press.

Reynolds, C. S., V. Huszar, C. Kruk, L. Naselli-Flores, and S. Melo. 2002. Towards a functional classification of the freshwater phytoplankton. J. Plankton Res. **24**: 417–428. doi:10.1093/plankt/24.5.417

Roegner, A., and others. 2020. Harmful algal blooms threaten the health of peri-urban fisher communities: A case study in Kisumu Bay, Lake Victoria, Kenya. Expo Health **12**: 835–848. doi:10.1007/s12403-019-00342-8

Rowe, M. D., and others. 2016. Vertical distribution of buoyant *Microcystis* blooms in a Lagrangian particle tracking model for short-term forecasts in Lake Erie. J. Geophys. Res. Oceans **121**: 5296–5314. doi:10.1002/2016JC011720

Ryabov, A., O. Kerimoglu, E. Litchman, I. Olenina, L. Roselli, A. Basset, E. Stanca, and B. Blasius. 2021. Shape matters: the relationship between cell geometry and diversity in phytoplankton. Ecol. Lett. **24**: 847–861.

Rynearson, T. A., and E. V. Armbrust. 2005. Maintenance of clonal diversity during a spring bloom of the centric diatom *Ditylum brightwellii*. Mol. Ecol. **14**: 1631–1640. doi:10.1111/j.1365-294X.2005.02526.x

Schaeffer, B. A., D. Kamykowski, L. McKay, G. Sinclair, and E. J. Milligan. 2007. A comparison of photoresponse among ten different *Karenia brevis* (Dinophyceae) isolates. J. Phycol. **43**: 702–714. doi:10.1111/j.1529-8817.2007.00377.x

Schwaderer, A. S., K. Yoshiyama, P. de Tezanos Pinto, N. G. Swenson, C. A. Klausmeier, and E. Litchman. 2011. Eco-evolutionary patterns in light utilization traits and distributions of freshwater phytoplankton. Limnol. Oceanogr. **56**: 589–598. doi:10.4319/lo.2011.56.2.0589

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Segura, A. M., C. Kruk, D. Calliari, and H. Fort. 2013. Use of a morphology-based functional approach to model phytoplankton community succession in a shallow subtropical lake. Freshw. Biol. **58**: 504–512. doi:10.1111/j.1365-2427.2012.02867.x

Seto, D. S., L. Karp-Boss, and M. L. Wells. 2019. Effects of increasing temperature and acidification on the growth and competitive success of *Alexandrium catenella* from the Gulf of Maine. Harmful Algae **89**: 101670. doi:10.1016/j.hal.2019.101670

Shapiro, J. 1984. Blue-green dominance in lakes: The role and management significance of pH and CO_2. Int. Revue Ges. Hydrobiol. **69**: 765–780. doi:10.1002/iroh.19840690602

Simis, S. G. H., S. W. M. Peters, and H. J. Gons. 2005. Remote sensing of the cyanobacterial pigment phycocyanin in turbid inland water. Limnol. Oceanogr. **50**: 237–245. doi:10.4319/lo.2005.50.1.0237

Smayda, T. J. 1997. Harmful algal blooms: Their ecophysiology and general relevance to phytoplankton blooms in the sea. Limnol. Oceanogr. **42**: 1137–1153. doi:10.4319/lo.1997.42.5_part_2.1137

Smayda, T. J., and C. S. Reynolds. 2001. Community assembly in marine phytoplankton: Application of recent models to harmful dinoflagellate blooms. J. Plankton Res. **23**: 447–461. doi:10.1093/plankt/23.5.447

Smith, V. H. 1983. Low nitrogen to phosphorus ratios favor dominance by blue-green algae in lake phytoplankton. Science **221**: 669–671. doi:10.1126/science.221.4611.669

Ståhl-Delbanco, A., L.-A. Hansson, and M. Gyllström. 2003. Recruitment of resting stages may induce blooms of *Microcystis* at low N:P ratios. J. Plankton Res. **25**: 1099–1106. doi:10.1093/plankt/25.9.1099

Steffen, M. M., and others. 2017. Ecophysiological examination of the Lake Erie *Microcystis* bloom in 2014: Linkages between biology and the water supply shutdown of Toledo, OH. Environ. Sci. Technol. **51**: 6745–6755. doi:10.1021/acs.est.7b00856

Sterner, R. W., K. L. Reinl, B. M. Lafrancois, S. Brovold, and T. R. Miller. 2020. A first assessment of cyanobacterial blooms in oligotrophic Lake Superior. Limnol. Oceanogr. **65**: 2984–2998. doi:10.1002/lno.11569

Strzepek, R. F., B. L. Nunn, L. T. Bach, J. A. Berges, E. B. Young, and P. W. Boyd. 2022. The ongoing need for rates: Can physiology and omics come together to co-design the measurements needed to understand complex ocean biogeochemistry? J. Plankton Res. **44**: 485–495. doi:10.1093/plankt/fbac026

Stumpf, R. P. 2021, *2021 Lake Erie algal bloom more severe than predicted by seasonal forecast*. NOAA.

Tahvanainen, P., T. J. Alpermann, R. I. Figueroa, U. John, P. Hakanen, S. Nagai, J. Blomster, and A. Kremp. 2012. Patterns of post-glacial genetic differentiation in marginal populations of a marine microalga. PLoS One **7**: e53602. doi:10.1371/journal.pone.0053602

Tee, H. S., S. A. Wood, K. Bouma-Gregson, G. Lear, and K. M. Handley. 2021. Genome streamlining, plasticity, and metabolic versatility distinguish co-occurring toxic and non-toxic cyanobacterial strains of *Microcoleus*. Mbio **12**: e0223521.

Thomas, M. K., C. T. Kremer, C. A. Klausmeier, and E. Litchman. 2012. A global pattern of thermal adaptation in marine phytoplankton. Science **338**: 1085–1088. doi:10.1126/science.1224836

Thomas, M. K., C. T. Kremer, and E. Litchman. 2016. Environment and evolutionary history determine the global biogeography of phytoplankton temperature traits. Global Ecol. Biogeogr. **25**: 75–86. doi:10.1111/geb.12387

Thomas, M. K., and E. Litchman. 2016. Effects of temperature and nitrogen availability on the growth of invasive and native cyanobacteria. Hydrobiologia **763**: 357–369. doi:10.1007/s10750-015-2390-2

Thorson, J. T., and C. Minto. 2014. Mixed effects: A unifying framework for statistical modelling in fisheries biology. ICES J. Mar. Sci. **72**: 1245–1256.

Tilman, D. 1990. Constraints and tradeoffs: Toward a predictive theory of competition and succession. Oikos **58**: 3–15. doi:10.2307/3565355

Van de Waal, D. B., L. T. Bach, E. Berdalet, K. M. Brandenburg, S. Suikkanen, S. Wohlrab, P. J. Hansen, and A. Kremp. 2022. HABs under global change: Experimental conditions and approaches, p. 36–63. *In* M. L. Wells, M. Burford, A. Kremp, M. Montresor, and G. C. Pitcher [eds.], *Guidelines for the study of climate change effects on HABs*. UNESCO-IOC/SCOR.

Vesk, P. A., W. K. Morris, W. C. Neal, K. Mokany, and L. J. Pollock. 2021. Transferability of trait-based species distribution models. Ecography **44**: 134–147. doi:10.1111/ecog.05179

Violle, C., M.-L. Navas, D. Vile, E. Kazakou, C. Fortunel, I. Hummel, and E. Garnier. 2007. Let the concept of trait be functional! Oikos **116**: 882–892. doi:10.1111/j.0030-1299.2007.15559.x

Violle, C., B. J. Enquist, B. J. McGill, L. Jiang, C. H. Albert, C. Hulshof, V. Jung, and J. Messier. 2012. The return of the variance: Intraspecific variability in community ecology. Trends Ecol. Evol. **27**: 244–252. doi:10.1016/j.tree.2011.11.014

Walker, T. W. N., and others. 2022. Functional traits 2.0: The power of the metabolome for ecology. J. Ecol. **110**: 4–20. doi:10.1111/1365-2745.13826

Wang, G., Z. Lee, D. R. Mishra, and R. Ma. 2016. Retrieving absorption coefficients of multiple phytoplankton pigments from hyperspectral remote sensing reflectance measured over cyanobacteria bloom waters. Limnol. Oceanogr. Methods **14**: 432–447. doi:10.1002/lom3.10102

Wang, J., S. C. James, J. A. Back, and J. T. Scott. 2022. Incorporating parameter variability into Monod models of nutrient-limited growth of non-diazotrophic and

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

diazotrophic cyanobacteria. Environ. Microbiol. doi:10.1111/1462-2920.16188

Wasmund, N. 1997. Occurrence of cyanobacterial blooms in the Baltic Sea in relation to environmental conditions. Int. Rev. Ges. Hydrobiol. Hydrogr. **82**: 169–184. doi:10.1002/iroh.19970820205

Weithoff, G., and B. E. Beisner. 2019. Measures and approaches in trait-based phytoplankton community ecology—From freshwater to marine ecosystems. Front. Mar. Sci.: 6: 40.

Wells, M. L., and others. 2020. Future HAB science: Directions and challenges in a changing climate. Harmful Algae **91**: 101632. doi:10.1016/j.hal.2019.101632

Wilkinson, G. M., J. A. Walter, C. D. Buelo, and M. L. Pace. 2022. No evidence of widespread algal bloom intensification in hundreds of lakes. Front. Ecol. Environ. **20**: 16–21. doi:10.1002/fee.2421

Willis, A., A. W. Chuang, J. N. Woodhouse, B. A. Neilan, and M. A. Burford. 2016. Intraspecific variation in growth, morphology and toxin quotas for the cyanobacterium, *Cylindrospermopsis raciborskii*. Toxicon **119**: 307–310. doi:10.1016/j.toxicon.2016.07.005

Wilson, A. E., O. Sarnelle, B. A. Neilan, T. P. Salmon, M. M. Gehringer, and M. E. Hay. 2005. Genetic variation of the bloom-forming cyanobacterium *Microcystis aeruginosa* within and among lakes: Implications for harmful algal blooms. Appl. Environ. Microbiol. **71**: 6126–6133. doi:10.1128/AEM.71.10.6126-6133.2005

Wohlrab, S., E. Selander, and U. John. 2017. Predator cues reduce intraspecific trait variability in a marine dinoflagellate. BMC Ecol. **17**: 8.

Wood, S. A., and others. 2020. Toxic benthic freshwater cyanobacterial proliferations: Challenges and solutions for enhancing knowledge and improving monitoring and mitigation. Freshw. Biol. **65**: 1824–1842. doi:10.1111/fwb.13532

Work, K. A., and K. E. Havens. 2003. Zooplankton grazing on bacteria and cyanobacteria in a eutrophic lake. J. Plankton Res. **25**: 1301–1306. doi:10.1093/plankt/fbg092

Wright, I. J., and others. 2004. The worldwide leaf economics spectrum. Nature **428**: 821–827. doi:10.1038/nature02403

Wyman, M., and P. Fay. 1986. Interaction between light quality and nitrogen availability in the differentiation of akinetes in the planktonic cyanobacterium *Gloeotrichia echinulata*. Br. Phycol. J. **21**: 147–153.

Yang, Z., and F. Kong. 2012. Formation of large colonies: A defense mechanism of *Microcystis aeruginosa*under continuous grazing pressure by flagellate *Ochromonas* sp. J. Limnol. **71**: e5.

Xiao, M., M. P. Adams, A. Willis, M. A. Burford, and K. R. O'Brien. 2017. Variation within and between cyanobacterial species and strains affects competition: Implications for phytoplankton modelling. Harmful Algae **69**: 38–47. doi:10.1016/j.hal.2017.10.001

Zhang, S., T. Zheng, N. Lundholm, X. Huang, X. Jiang, A. Li, and Y. Li. 2021. Chemical and morphological defenses of *Pseudo-nitzschia multiseries* in response to zooplankton grazing. Harmful Algae **104**: 102033.

Zohary, T., and R. D. Robarts. 1990. Hyperscums and the population dynamics of *Microcystis aeruginosa*. J. Plankton Res. **12**: 423–432. doi:10.1093/plankt/12.2.423

Acknowledgments

This work was in part supported by the NSF grants 16-38958 and 17-54250, and the MSU Foundation. Chris Klausmeier helped with Fig. 4. This is Kellogg Biological Station contribution number 2323.

Submitted 27 April 2022
Revised 09 September 2022
Accepted 03 November 2022

23782242, 2023, 2, Downloaded from https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lo2.10294 by University Of Wisconsin @ Green Bay, Wiley Online Library on [17/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License